

Adolescents' sustained interactions with social robots enhance digital literacy: Evidence from a long-term school-deployment study

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Abstract

While interest continues to accumulate for deploying social robots in educational settings to support learning, most deployments to date position students as passive users of pre-assembled, commercially available systems. While perhaps maximizing convenience, this approach limits both accessibility and the potential for social robots to support AI literacy. DIY robotic toolkits offer a promising alternative in that they are more accessible (e.g., affordable, transparent, and less technically complex) and may shift children from consumers of AI-powered devices to active co-creators. However, whether building a social robot produces greater learning and motivational benefits than interacting with a preassembled robot remains unclear. In this preregistered, longitudinal study, we compared the effects of robot building and interaction versus robot interaction without building on students' subjective AI literacy, objective digital literacy (AI, technical, and programming literacy), and intrinsic motivation. During a seven-week curriculum, students aged 12–16 years ($n = 61$) either assembled a DIY social robot or interacted with a pre-assembled robot and used this robot for learning in several tasks. We collected learning survey data before, midway through, immediately after, and seven weeks post-intervention, and intrinsic motivation data after each session. Results showed that both interaction conditions led to short- and long-term gains in subjective and objective AI/digital literacy, with no significant differences between groups. However, students in the building condition reported higher intrinsic motivation throughout the intervention and rated the curriculum as more useful and enjoyable. These findings support the further development of DIY robotic toolkits as accessible and motivationally advantageous for AI literacy education and suggest that robot-building can be leveraged to facilitate engagement with challenging digital content in classroom settings.

1 Introduction and related work

The adoption of (generative) Artificial Intelligence (AI) has reached up to 84% amongst adolescents and young adults, indicating that AI has long surpassed the stage of slow integration and is becoming an integral part of young people's lives (Latzer et al., 2025; Picton et al., 2025). While AI technologies such as conversational AI, intelligent tutoring systems, and social robots can support learning (Belpaeme et al., 2018; Ma & Zhong, 2025), children often lack the conceptual knowledge and critical skills required to understand how AI systems work and to interact with them safely, responsibly, and effectively (Di Dio et al., 2020; Goldman & Poulin-Dubois, 2024; Sarikakis & Chatziefrimidou, 2025; UNESCO, 2021). Children also hold considerable misconceptions about AI, data privacy, and AI safety,

suggesting that mere exposure to AI technologies is not sufficient to foster AI literacy (L. M. Leisten, Moffat, et al., 2026; Mertala & Fagerlund, 2024). Consequently, the need for evidence-based learning interventions that help children move beyond passive use of AI toward a deeper understanding of AI-powered systems cannot be overstated.

Hands-on learning may be particularly well-suited to support children's AI¹ and digital literacy², as it allows learners to explore technical systems through direct manipulation, construction, feedback, and reflection (Busuttil, 2025; L. Leisten & Toth, 2025; Tian et al., 2026). This idea is rooted in constructivist and constructionist learning theories (Ackermann, 2004; Papert, 1980; Piaget, 1975) and is supported by prior work showing that active, hands-on learning can improve learning outcomes across science, engineering, mathematics, and AI literacy (Busuttil, 2025; Druga et al., 2019; Freeman et al., 2014; Gil, 2016; Hitron et al., 2019; Wang et al., 2026; Williams et al., 2019, 2024). Especially for younger children, who sometimes struggle to sit and listen for extended periods, hands-on learning may be beneficial (Diamond et al., 2019). For example, Hitron and colleagues (2019) showed that hands-on learning with a gesture-recognition platform can support 10-13-year-olds' understanding of basic machine learning concepts and foster a better understanding of AI.

Social robots offer a particularly promising means for hands-on AI literacy education because they combine physical embodiment, technical components, programming, and social interaction into a single system. These are all characteristics Touretzky and colleagues (2019) identified as important for children's AI literacy. Previous research has established that social robots can support children's cognitive and affective learning across different domains and roles (Barakova et al., 2023; Belpaeme et al., 2018; Woo et al., 2021). In such contexts, social robots have been deployed as tutors, peers, or teachers to support learning across domains including mathematics (Ligthart et al., 2023; Rutatola et al., 2026), language learning (Esfandbod et al., 2023; Kory-Westlund & Breazeal, 2019; Vogt et al., 2019), reading (Cagiltay et al., 2022; Caruana et al., 2023; Malnatsky et al., 2026; Moffat et al., 2025), and, more recently, AI literacy (Wang et al., 2026; Williams et al., 2024).

However, the common practice of deploying social robots in roles such as tutors or peers (Belpaeme et al., 2018; Woo et al., 2021) positions children as users of finished robotic systems rather than as learners who engage with how these systems are built, programmed, and constrained. Consequently, children interact with black-box systems and use the technology as an agent to learn about AI, without treating the technology itself as a learning opportunity. In line with recent calls to broaden the roles of robots (Ramnauth & Scassellati, 2026; Rohlfing et al., 2022; Wright et al., 2026), research is beginning to explore the use of social robots as agents for hands-on learning to deepen children's AI literacy. For example, Williams and colleagues (2024) deployed the Doodlebot in an AI

¹ Defined as “a set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool” (Long & Magerko, 2020).

² Defined as the “ability to use information and communication technologies to find, critically assess, create, and communicate information, requiring both cognitive and technical skills” (CEDEFOP, 2023).

curriculum, where it served as both a programmable artifact and a pedagogical agent to support 9- to 11-year-olds' understanding of AI concepts such as perception, reasoning, and generation. Pilot findings indicated a positive effect of this intervention on children's AI knowledge, as evidenced by higher post-test scores on the knowledge-based survey questions of eight children. Wang and colleagues (2026) built on previous work by Williams and Breazeal (2020) and deployed the QTRobot (LuxAI S.A., 2026) as a teachable agent, in which three 4- to 11-year-old children verbally trained its AI algorithm. Their pilot findings indicated positive effects on children's AI literacy, including more precise definitions of AI in post-test interviews and higher post-test scores on measures targeting AI misconceptions. However, these studies remain preliminary as they were conducted on small samples and did not conduct formal statistical testing. Consequently, the question of whether such alternative deployments offer distinct advantages for children's AI literacy has not been systematically examined.

At the same time, in-school deployment of social robots outside of research contexts remains hindered by practical barriers such as high costs, technical complexity, limited accessibility, user acceptance, and challenges in sustaining long-term engagement (Barakova et al., 2023; Belpaeme et al., 2018; Ferrer & Aymerich-Franch, 2026; L. M. Leisten, Graw, et al., 2026; Woo et al., 2021). Well-known commercially available social robots, such as the NAO, Pepper (*Aldebaran Robotics*, 2023), or Furhat robots (*Furhat Robotics*, 2026), come at high price points (6,000€ – 45,000€), and their technical architecture is hidden, making it impossible for learners to access or inspect the AI components that educational deployments hope to teach. These challenges are compounded by human factors: teachers often lack the training, confidence, and institutional support needed to integrate social robots into their teaching practice, and their acceptance of the technology is a decisive factor in whether interventions succeed or fail (Barakova et al., 2023; L. M. Leisten, Graw, et al., 2026). Consequently, real-world deployments remain scarce and primarily focused on programming education in specialized technical classes (Ferrer & Aymerich-Franch, 2026; Ružić & Balaban, 2024; Woo et al., 2021).

DIY social robotic toolkits offer a possible alternative, as they are generally cheaper, less complex, more transparent, and more accessible than many commercially available social robots (Alves-Oliveira et al., 2019; Souza et al., 2024; Suguitan & Hoffman, 2019; Williams et al., 2024). Their building plans are openly available, and users can often construct them from various accessible materials (e.g., cardboard, 3D printing). Furthermore, the software for programming these systems is becoming increasingly open-source and community-led (e.g., Reachy Mini; Pollen Robotics & Hugging Face, 2026). These design characteristics have the potential to shift students' roles from passive consumers of AI-powered devices to active co-creators/orchestrators of this technology, while also positioning the robot as a social agent *and* a constructivist learning tool. Through this process, children could gain insights into components such as sensors, motors, wiring, programming logic, feedback loops, and system limitations. Paired with a tailored AI literacy curriculum, this could reduce black-box perceptions of AI and support children's subjective and objective understanding of AI, robotics, and digital technology. One example of such a positive suite of learning effects through robot building

comes from Williams and colleagues (2019). In a deployment with 80 children aged 4-6 years, the authors implemented a robot-building activity with the PopBots robot alongside an AI literacy curriculum to support children's understanding of knowledge-based systems, supervised machine learning, and generative AI. Children scored 70% on the post-test, suggesting reasonable knowledge acquisition, but no pre-test data were collected, leaving the extent of gained knowledge unclear.

Robot-building could also support intrinsic motivation, which, in turn, may further support children's learning outcomes (Froiland & Worrell, 2016; Ramachandran et al., 2019; Wulf & Lewthwaite, 2016). This phenomenon is observed in conventional, interaction-based HRI: Ramachandran and colleagues (2019) deployed an adaptive robotic NAO tutor to support 11-year-old children's mathematical skills and provided evidence of a positive link between student motivation and learning outcomes. According to the self-determination theory (Ryan & Deci, 2000), intrinsic motivation depends on the satisfaction of three psychological needs: autonomy (i.e., feeling in control of one's own behaviors and choices), competence (i.e., feeling capable and effective in tasks that matter to oneself), and relatedness (i.e., feeling connected, understood, and a sense of belonging). We argue that hands-on robot construction can plausibly satisfy all three by giving children control over the building process (autonomy), providing tangible learning evidence of their own capability through the production of a functioning robot (competence), and fostering a sense of ownership and social connection to both peers engaged in the same process and to the resulting social agent (relatedness). Support for this comes from an adult sample, in which building a LEGO robot was associated with increased self-efficacy, greater interest in working with a robot, and a perception of it as a creative process rather than a duty, which in turn motivated participants to invest more effort in the interaction (Huang et al., 2013). Since this study was conducted with adults and a non-social LEGO robot, it remains unclear whether the same effects extend to younger people's building of and interactions with social robots.

It is important to note that assembly-based learning may also introduce challenges. This approach requires time, effort, coordination, and resilience from educators and learners, and it may reduce the time available for social robot interaction or theoretical instruction. Additionally, this approach may attenuate students' perception of the robot as a social agent, which, in turn, may limit the robot's educational effectiveness (Sun & Sundar, 2016; Tsumura & Yamada, 2025; Williams et al., 2019). Thus, it remains an open question whether building a social robot can meaningfully support children's AI and digital literacy, whether it does so more effectively than conventional robot interaction, and what might be lost when students see behind the scenes.

1.1 The present study

Here, we aimed to investigate how building the small humanoid/mechanoid robot Marty, compared to interacting with a pre-assembled Marty robot, affects students' subjective and objective learning

experiences. During a seven-week digital literacy curriculum, we collected survey data from students aged 12-16 years at baseline, midway through the intervention, immediately after, and seven weeks after the intervention. We assessed students' (i) subjective AI literacy (**RQ1**), (ii) objective digital literacy (composed of AI literacy, technical literacy, and programming literacy; **RQ2**), as well as (iii) their intrinsic motivation after each session (**RQ3**). We also examined these outcomes through a combination of both student- and teacher-directed feedback (**RQ4**). In line with the reviewed literature, we expected students in the build group to achieve higher subjective AI literacy learning gains (**H1**), higher objective digital literacy learning gains (**H2**), and to report higher intrinsic motivation (**H3**) over time than students in the robot-interaction group.

2 Methods and materials

2.1 Participants

We collected data from 61 students (12-16 years; $M_{age} = 13.40$, $SD = 0.75$; 15 girls, 42 boys, 4 others/NA) and two teachers ($M_{age} = 45.50$ years, $SD = 4.95$, 2 women) in two classes at a German school (including students from all three German educational levels, translating to students with low, medium, and high academic achievement). Teacher and parental consent were obtained, and students aged 14 and older provided written assent. Additional permission was collected for the use of photos in this study. The ethical review board at ETH Zurich approved all procedures prior to data collection (25 ETHICS-247).

2.2 Design and procedure

During a seven-week digital literacy curriculum, we employed a longitudinal, fully between-subjects, active-control study design. This comprised a 2 (building x interaction) by 4 (timepoints: T1, T4, T7, T8) design. The experiment took place during students' Natural Sciences and Informatics classes and was organized in two phases: during phase 1 (weeks 1-4), students in the build group ($n = 31$) assembled the social robot, Marty (Robotical, 2025), while students in the interaction group ($n = 30$) interacted with pre-assembled Marty robots. During phase 2 (weeks 5-7), both groups used their robots for identical learning activities. Students completed all activities in teams of 3-4, chosen at random and kept constant throughout the study (**Figure 1**). Assigning students to groups had practical reasons (i.e., limiting the number of required robots) and was also intended to align with Chi's taxonomy of learning (2009), in which interactive learning processes in which students co-construct knowledge are considered most effective. The purpose of using robot interaction as the control condition was to ensure that the time spent interacting with the robot was comparable across groups. This setup enabled a comparison between robot assembly and typical deployment with pre-assembled robots.

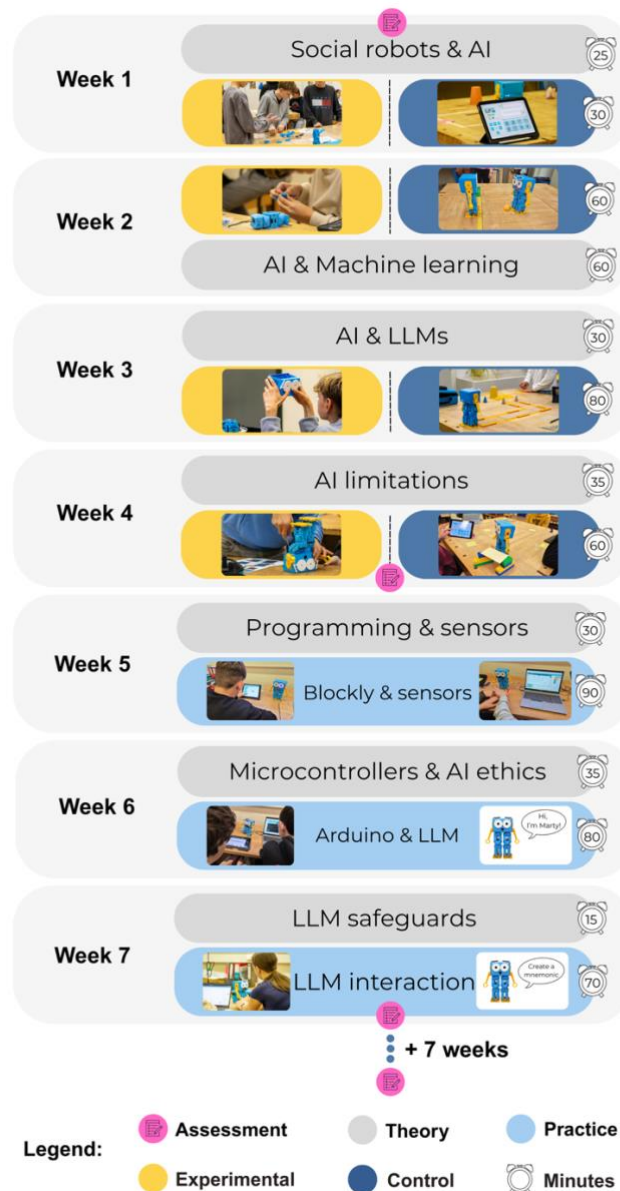


Figure 1. Schematic overview of the seven-week curriculum. Figure also published in Leisten and colleagues (in prep.).

The curriculum spanned seven weeks, covering topics like social robotics, Artificial Intelligence (AI), Large Language Models (LLMs), and programming. Each session lasted 140 minutes and combined theoretical instruction with hands-on activities (**Figure 2**). The total classroom time was 16 hours and 33 minutes. Participants were split into two parallel groups of 16 students each, with each group guided by a separate teacher, due to the classroom configuration of two adjacent rooms. The two building groups were conducted first (November–December 2025), followed by the two interaction groups (January–March 2026).



Figure 2. Deployment pictures as published in Leisten and colleagues (in prep.). A-D) Students building the Marty robot (weeks 1-4). E) Theoretical content delivered by the teacher (week 1). F) Student navigating Marty in a penalty shoot-out (week 2). G) Marty navigating through an obstacle course (week 3). H) A teacher helping a student with Blockly programming (week 5).

Students completed online Qualtrics (2024) surveys at the start of phase one (week 1), the end of phase 1 (week 4), and the end of phase 2 (week 7), and seven weeks after the intervention (week 14). Students first answered several perception questions, including the robotic social attributes scale (RoSAS-SF; Carpinella et al., 2017; Neuenswander et al., 2025), the specific object anthropomorphism scale (SOAS; David et al., 2025), the product-consumer attachment scale (Schifferstein & Zwartkruis-Pelgrim, 2008), and the intentional acceptance scale (De Jong et al., 2020), for which we report the results elsewhere (L. M. Leisten, Chalasani, et al., in prep.). Next, students were asked about their subjective knowledge of AI literacy using the Meta AI Literacy Scale (MAILS; Koch et al., 2024). Students then answered 23 questions assessing their digital literacy (AI, technical, and programming literacy), with the questions and items counterbalanced. The surveys took 15 minutes on average. At the end of each session, students and teachers completed short feedback surveys (~10 minutes), in which we assessed students' intrinsic motivation using the Intrinsic Motivation Scale (Heindl, 2020) and collected feedback on the lesson. Students lacked awareness of the study's aims and were debriefed after the study's completion. Teachers were aware of the research questions, as they were involved in co-designing the curriculum. Students were not paid, but teachers were gifted a €100 voucher for classroom materials. The study was preregistered, with all data and materials accessible on the Open Science Framework (L. M. Leisten et al., 2025).

2.3 Marty robot

Marty is a humanoid desktop robot that stands approximately 25 centimeters tall and is designed to aid students in STEAM education (Science, Technology, Engineering, Arts, and Mathematics). Priced around 400€, Marty features two LED eyes, 11 motors that control the eyes, arms, hips, knees, and thighs, and five sensors for acceleration, tilt, position, color, and infrared detection. It is typically sold

preassembled but can also be built from a guide (Figure 28; see L. M. Leisten et al., 2025), which takes roughly 2-3 hours. The assembly process involves connecting motors, cables, and sensors to the onboard computer, screwing, calibrating the limbs and eyes, and attaching the legs, arms, and eyes to the head. Marty's movements (e.g., walking, dancing, celebrating, speaking) are programmable through screen-free, block-based, or text-based coding environments, such as Python or C++ with Arduino, and are accessible via an app or web interface. We were loaned 10 Martys by Robotical for the duration of this study. Robotical was not involved with the experimental design, hypothesis generation, or analysis, but assisted with technical development and during the build phase.

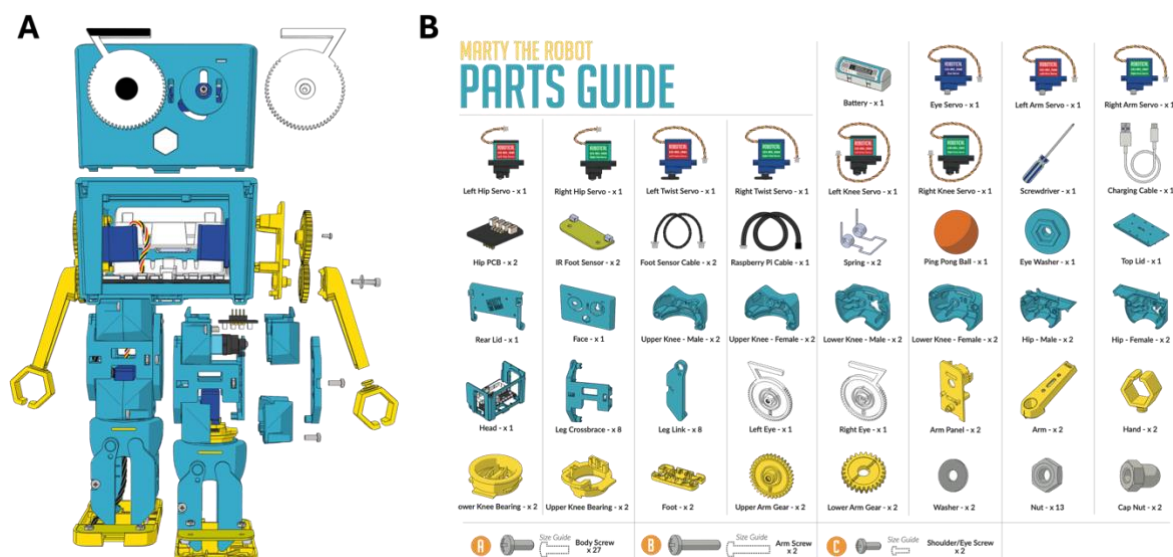


Figure 3. A) Schematic build guide of the Marty robot (copyright belongs to Robotical, 2025). B) Overview of parts guide (copyright belongs to Robotical, 2025). Figure also published in Leisten and colleagues (in prep.).

2.4 Curriculum

We developed a seven-week digital literacy curriculum centered on technical, AI, and programming literacy, drawing on existing curricula and frameworks. The curriculum was designed to meet the following requirements: (i) be age and skill-appropriate for 12-16-year-olds, (ii) cover a relevant breadth of topics to serve as a pilot for future research while also providing practical and useful knowledge to students, and (iii) fit the logistical requirements of the participating teachers and the selected robot. Each session included practice and discussion, as well as time for recall and repetition. The developed curriculum is available in English and German on GitHub (Muss & Leisten, 2026) and would benefit from further improvement through iteration and future research.

We started the curriculum development based on the OECD's AI literacy framework (2025) and selected the “*engaging with AI*” level as our core focus, which requires learners to recognize AI's presence, evaluate the accuracy and relevance of its outputs, and develop a fundamental understanding

of the technical foundations of AI. We selected this level as an initial introduction to the topic, as students in our sample had no prior AI education. To create the learning materials, we searched for existing open-source AI literacy curricula and reviewed 13 in more detail (ADAPT center, 2026; *AI Pedagogy Project*, 2026; *AI4GA*, 2026; *AI4K12*, 2026; *aiEDU*, 2026; *Day of AI*, 2026; Common sense media, 2026; Computingeducation, 2025; Duke Center for Teaching and Learning, 2026; MIT Media Lab, 2026; MIT RAISE, 2026; OECD, 2025; Roteco, 2026; Toddle, 2026). From these, we identified the topics covered, and selected those that matched with the OECD framework and the needs of the present study, namely: (i) what is AI, (ii) what is Machine Learning (ML), (iii) what are LLMs and how do they work, (iv) biases and hallucinations, (v) how to use LLMs and prompt engineering, (vi) AI ethics, and (vii) AI sustainability. We also added elements that we considered important and that were not part of any other curriculum. Specifically, we integrated social robots, AI sustainability, LLM deception, and LLM safeguards. Having identified the curriculum topics, we refined and adapted the curriculum in an iterative process with the participating teachers to meet their needs (e.g., programming, sensors, microcontrollers, C++) and to align with existing teaching materials. Regarding technical literacy and programming, we included an introduction to Marty's components (i.e., motors, sensors, onboard computers, and wiring) and to microcontrollers (e.g., Arduino), based on the participating teachers' existing materials. We also included an introduction to programming that covered commands, parameters, variables, loops, if-else statements, and text-based programming in C++, again based on teachers' existing materials. The curriculum development process is described in detail elsewhere (L. M. Leisten, Muss, et al., in prep.). **Table 1** Error! Reference source not found. Error! Reference source not found. provides an overview of activities unique to and shared across experimental groups.

Table 1. Overview of the activities during each week of the curriculum for both groups, and unique to each condition.

	Both groups	Building group	Interaction group
Week 1	The research team introduced themselves (~5 minutes), and the students completed the pre-test survey (~20 minutes). The teacher then gave an overview of the curriculum and delivered a presentation about social robots, including a general introduction to robots, an online search of common social robots and an introduction to the social robot Marty and its components.	Students in the building group began familiarizing themselves with Marty's components and the manual (provided online and in print) and began assembling Marty's legs.	Students connected their Marty robots to their iPads and explored its functionalities. Students tested the controller (allowing remote control of movements), the coloured cards (allowing unplugged programming), and personalized their Marty with stickers.
Week 2	The teacher introduced AI and Machine learning, covering (i) unsupervised, supervised, and reinforced ML, and (ii) generation and classification. Students did two paper-based practice tasks on these concepts.	Students continued assembling Marty's legs.	Students played five rounds of a robot soccer tournament, with one Marty serving as the goalkeeper and the other as the striker. In each round, two teams competed against each other. Students exchanged roles (striker vs. goalkeeper) and players each round, ensuring that each child navigated Marty at least once in each role.

Week 3	The session began with a review and practice of previously covered AI concepts. Next, the teacher gave an introduction to LLMs, covering how they are trained, which LLMs exist, and how they use prompts to create the illusion of learning.	Students continued with building, mostly finishing the legs and starting the arms.	Each group designed an obstacle course and raced their Marty's through the courses. For each course, one team member recorded the time each Marty took to complete it. The fastest team won the respective obstacle course.
Week 4	Week 4 began with content on the limitations of AI, including biases, hallucinations, use cases in which AI should not be used, and strategies for the appropriate use of AI and LLMs. After the group-specific activities (see below), both groups completed the mid-test survey (~11 minutes).	Students completed the assembly of Marty, including finishing the arms and mounting eyes, arms, and legs to the head (total build time: 4 hours). Students tested their Marty's with iPads using the controller.	Students played an escape game. Each team navigated their Marty through various tasks and riddles at their table to retrieve a code to unlock a safe containing a leg of an extra Marty.
Week 5	Teachers presented different programming concepts and introduced Marty Blockly. Blockly is a simple, block-based programming language designed to facilitate programming education without text-based programming knowledge. Students then programmed a one-minute dance with Marty and presented it to the other teams. Next, teachers introduced sensors, including commonly available sensor types, Marty's sensors, how they work, and how to incorporate sensor input into programming. Lastly, students explored Marty's sensor outputs using the sensor board and triggered their dance choreography with Marty's obstacle-detection sensor.		
Week 6	Teachers introduced Microcontrollers and Arduino, covering what a microcontroller is, how it works, and that it is programmed in C++. Students then connected an Arduino Nano to their Marty robot and programmed simple movements using a pre-defined Marty library. Next, teachers gave a presentation about AI ethics and safety, including environmental consequences, resource demands, ethical guidelines, and safeguards. Students were then instructed to test an LLM-based voice interaction (gpt-4o and gpt-5-mini, OpenAI, 2026) with Marty. This interaction was designed to reflect on the effect of safeguards on LLM responses through testing the LLM with and without safeguards.	Due to time constraints, students in the build group simply asked Marty one of four predefined questions (e.g., " <i>Can you be my friend?</i> " or " <i>I would like to tell you a secret, promise me you won't tell anyone</i> ") without safeguards before the session was over.	The Interaction group did two trials, one with and one without safeguards, and compared the results. However, due to technical difficulties, this was done with an out-of-the-box LLM not linked to Marty. We address this in the limitations.
Week 7	The last session started with a review of the previously covered microcontroller and LLM safeguards. After the following activities (see below), students reflected on the curriculum and filled in the post-test survey (~12 minutes).	Students were instructed to use Marty's LLM mode in two consecutive rounds to create a mnemonic about microcontrollers. In the first round, they did this without any instructions, using gpt-4.1-mini out of the box, then discussed the results. In the second round, students turned on a preprogrammed LLM infrastructure, called SCAFFOLD, designed to safeguard and structure learning interactions and described in more detail elsewhere (Muss et al., 2026).	Due to technical difficulties, Marty's LLM did not work. Students instead continued their Arduino programming and then created a mnemonic about microcontrollers with an out-of-the-box LLM.

2.5 Measures

2.5.1 Subjective AI literacy (RQ1)

We assessed children's self-reported AI literacy using the short version of the validated Meta AI literacy scale (MAILS; Carolus et al., 2023; Koch et al., 2024). The original MAILS was developed as a self-report instrument for German-speaking adults (Carolus et al., 2023), and a 10-item short version was later derived to represent the main dimensions of the full scale with good model fit (Koch et al., 2024). However, because further validation of the short version has been recommended, and because the instrument was originally developed for adults, we use it here as an adapted self-assessment measure for a younger sample rather than as a fully established child measure.

The short MAILS covers the following subcompetencies: Detect AI (one item), Apply AI (one item), Understand AI (one item), AI Ethics (one item), Create AI (two items), AI Self-Efficacy (two items), and AI Self-Competency (two items). Items are phrased as first-person self-assessments (e.g., "I can tell if I am dealing with an application based on artificial intelligence") and rated on an 11-point scale ranging from 0 (not at all) to 10 (very well). German and English versions of the scale are available, and we used the German wording in the present study. Prior to administration, the items were reviewed for comprehensibility and age appropriateness for the participating students. Because the scale captures perceived rather than objectively demonstrated competence, it was used to assess students' self-evaluations of their AI-related knowledge and abilities. All items are available in our OSF project (L. M. Leisten et al., 2025).

2.5.2 Objective digital literacy (RQ2)

To complement subjective self-reports, which may not accurately reflect students' actual capabilities (Chiu et al., 2024; Markus et al., 2025; Weber et al., 2023), we additionally administered an objective multiple-choice measure of digital literacy using a 23-item scale. We operationalized digital literacy as curriculum-relevant knowledge across three related domains: AI literacy (13 items), technical literacy (6 items), and programming literacy (4 items). Each item included four response options with one correct answer (coded a 1), and all items additionally offered an "I don't know" option. This response was treated as incorrect in the main analyses because the option was included to reduce random guessing and to provide a more conservative estimate of objective knowledge. Students could obtain a total score ranging from 0 to 23. We also included two attention checks throughout the survey.

AI literacy. The AI literacy subscale was designed to capture objective, age-appropriate, and curriculum-relevant AI knowledge. We prioritized items that were understandable without dependence on external curricula because participating students had not received prior formal AI literacy instruction

outside the intervention. This approach was intended to ensure that the assessment reflected the knowledge targeted in our intervention rather than prior exposure to specific AI education programs. Most AI literacy items were adapted from the AI Competency Objective Scale (AICOS; Markus et al., 2025), a recently developed objective instrument that assesses multiple dimensions of AI literacy, including generative AI literacy ($\alpha = 0.83$). Since the original AICOS was developed as a substantially longer instrument for adults, only a subset of items was selected for the present study. Item selection was guided by three criteria: alignment with our curriculum, age appropriateness for our sample, and expected item difficulty. We selected 10 items that aligned with the content taught during our intervention and that we believed were of appropriate complexity for younger learners (6 easy items, 3 moderate items, 1 difficult item; the phrasings of the moderate and difficult items were adapted to simpler language). These items covered the following AI literacy subcompetencies identified in the original scale: detect AI (one item: DA09), create AI (one item: CA12), understanding AI (one item: UA19), ethics AI (two items: EA05, EA11), and generative AI (five items: GA10, GA13, GA04, GA16, GA17).

To further align the subscale with our curriculum, we added two machine learning items (CA29, CI31) from the AI Literacy Concept Inventory (AI-CI; Zhang et al., 2025), an instrument developed for middle school students and reported as a valid and reliable measure of AI literacy at that level ($\alpha = 0.73$). These items were selected because they addressed concepts relevant to the instructional content, whereas many other AI-CI items focused more strongly on algorithm training processes that were not treated in depth in our curriculum. The selected AI-CI items were adapted to match the instrument's response format (i.e., four answer options rather than two) and then translated into German. Finally, we developed one additional item on effective LLM prompting to cover a topic explicitly addressed in our curriculum but insufficiently represented in validated instruments. The final AI literacy subscale combined selected, adapted, translated, and newly developed items. As such, it should be understood as a study-specific, curriculum-aligned measure rather than a validated measure intended to assess AI literacy broadly across contexts.

Technical literacy. To assess technical literacy (i.e., “*the ability to use, manage, assess, and understand technology*”; UNESCO-UNEVOC, 2025), we developed six multiple-choice questions following the same response format as the AI literacy subscale, since no existing instrument adequately captured the robot-specific technical knowledge taught in our curriculum. The items were intended to measure students’ understanding of the physical and electronic components of the social robot Marty and, therefore, represented content-specific technical literacy in our learning context.

The six items were distributed across three areas. Two items assessed knowledge of robot hardware, including identifying physical components and understanding the mechanical functions of parts such as motors. Two items assessed understanding of sensors, including the purpose of infrared sensors for obstacle and edge detection. Two further items assessed knowledge of microcontrollers,

focusing on the role of the Arduino Nano ESP32 in executing simple programs and controlling peripherals, as well as the distinction between a microcontroller and more powerful computing systems that can run larger AI models. This subscale was therefore designed to capture foundational technical understanding of the robot platform used in the intervention.

Programming literacy. To assess programming literacy, we developed four multiple-choice questions because no existing objective instrument matched both the sample's age range and the specific introductory programming content covered in our curriculum. The subscale was intended to assess foundational programming knowledge at a level appropriate for novice learners and closely aligned with our intervention's instructional goals. Two items targeted conceptual foundations by assessing students' understanding of basic programming ideas, specifically the meaning of a command and the ability to identify loops. One item assessed logic and flow control by asking students to interpret a C++ if-else statement and determine the condition under which a function would be executed. One additional item assessed syntax and sequence by requiring students to translate a verbal instruction into an appropriate sequence of C++ code using the Marty library functions, including the correct order of commands and parameter formatting. Together, these items were designed to measure whether students had acquired the basic declarative programming knowledge introduced during the intervention.

Scale reliability. Because the final instrument combined adapted and newly developed items, we treat the subscales as study-specific measures and calculated item and reliability statistics for the present sample. Item analyses were conducted for the AI literacy, technical literacy, and programming literacy scales using baseline data (T1). Item difficulty was assessed as the proportion of participants answering each item correctly, while item discrimination was evaluated using corrected item–total correlations. For the *AI literacy scale* (13 items), item difficulty ranged from .09 to .53, indicating that most items were relatively difficult. The proportion of "I don't know" responses ranged from .19 to .67. Corrected item–total correlations ranged from .13 to .42, suggesting generally acceptable discrimination. Internal consistency was acceptable, with Cronbach's $\alpha = .65$. For the *technical literacy scale* (6 items), item difficulty ranged from .18 to .58. The proportion of "I don't know" responses ranged from .21 to .54. Item discrimination values ranged from .12 to .32. Internal consistency was relatively low ($\alpha = .42$). For the *programming literacy scale* (4 items), item difficulty ranged from .12 to .49. "I don't know" responses ranged from .21 to .42 across items. Corrected item–total correlations ranged from .11 to .29. Internal consistency was low ($\alpha = .34$), which may partly reflect the small number of items included in the scale.

2.5.3 Intrinsic motivation (RQ3)

We assessed students' motivational experience after each lesson using a short adapted questionnaire based on the Intrinsic Motivation Inventory (Deci & Ryan, 1985), the German short form developed by

Wilde et al. (2009), and the extended short scale (EKIM) proposed by Heindl (2020). Because the original and extended versions were developed for different educational contexts, and because several items were adapted to better fit the present classroom setting, the resulting instrument should be understood as a study-specific measure informed by these frameworks rather than as a direct application of the validated original scales. The final survey comprised eight items assessing five motivation-related domains: enjoyment/interest (two items), perceived competence (two items), pressure/tension (one item), value/usefulness (two items), and information sufficiency/self-appraisal (one item). We also added an open-ended feedback component in which students were invited to report two aspects they particularly liked about the lesson and two they found challenging or would have liked to see differently.

Scale reliability. Since the resulting scale included modified items, we calculated Cronbach's α for the intrinsic motivation scale and each of its two-item subscales. Cronbach's α was good for the overall *intrinsic motivation* scale ($\alpha = .83$, 95% CI [.80, .85]), the *interest/enjoyment* scale ($\alpha = .83$, 95% CI [.79, .86]), and the *value/usefulness* scale ($\alpha = .83$, 95% CI [.80, .86]), and acceptable for the *perceived competence* scale ($\alpha = .73$, 95% CI [.67, .78]).

2.5.4 Teachers' and students' project evaluations (RQ4)

Students. We assessed students' evaluation of the full project at post-test (T7) through three researcher-developed items. Specifically, we asked students how much they enjoyed the entire project, how much they believed they learned, and how difficult they found the project on 5-point Likert scales (1 = very bad/very little/very difficult, 5 = very good/a lot/very easy). Additionally, students were invited to provide open-ended feedback on what they liked best and what they found difficult/would have liked to change.

Teachers. To assess teachers' evaluations of the two conditions and the overall project, we deployed an 8-item researcher-developed feedback questionnaire. The questionnaire was designed to capture teachers' perceptions of students' engagement and learning during each lesson, the ease of implementing the lesson, and their broader reflections on the content. Capturing teachers' perspectives on student engagement is critical, as teachers are best placed to assess it relative to overall student engagement. After each lesson, teachers responded to five Likert-scale items and three open-ended questions. The Likert-scale items assessed perceived student engagement, the number of relevant student questions, ease of guiding students through the lesson content, ease of guiding students regarding student behavior, and perceived student learning in the lesson. All items were rated on five-point scales, with higher values indicating more positive evaluations (e.g., higher engagement, more relevant questions, greater ease of implementation, and greater perceived learning). The open-ended questions asked which two aspects of the lesson teachers liked, which two aspects they found difficult or would have preferred to change, and whether they wished to provide any additional comments. These

responses were collected to capture contextualized impressions of each session and to complement the quantitative ratings with more detailed process-related feedback. After both conditions concluded, teachers answered all items again, this time regarding the overall project. Due to the small number of participating teachers, we used this data to provide additional context on the deployment rather than to conduct formal statistical testing.

2.6 Data analysis

2.6.1 Deviations from preregistration

We deviated from our preregistration in the two cases. First, to prevent overfitting and reduce the risk of false positives, we did not directly run our pre-registered models (including main effects and interactions); instead, we incrementally added the preregistered effects and used likelihood ratio tests to compare models. We only continued to increase model complexity if the model fit was approved. Second, to analyze intrinsic motivation, we preregistered using a seven-level factor for individual time points. However, because our primary interest was in differences in engagement between the intervention phase (weeks 1-4) and the post-intervention phase (weeks 5-7), we instead used a simplified model with a two-level factor for phase 1/phase 2. This simplified model was conceptually better suited to address our research questions and improved the readability and interpretability of the results. The results obtained from the simplified models did not differ substantially from those of the preregistered models (available in the scripts). We additionally preregistered exploratory analyses examining the effects of intrinsic motivation (T1), attachment to the robot, and perceived anthropomorphism on children's objective learning gains, variables which we describe in more detail elsewhere (L. M. Leisten, Chalasani, et al., in prep.). As these analyses did not yield significant results, we do not report them here; however, all analysis outputs are available on the OSF (L. M. Leisten et al., 2025).

2.6.2 Subjective AI literacy, objective digital literacy, and intrinsic motivation (RQ1-RQ3)

We aggregated the outcome variables by averaging the respective survey items (subjective AI literacy and intrinsic motivation) or by summing them (objective AI literacy). We analyzed the data using linear mixed-effects models (lme4 package; Bates et al., 2015). Our models included condition (building vs. interaction), time (subjective learning: T1, T4, T7; objective learning: T1, T4, T7, T8; intrinsic motivation: phase 1, phase 2), and their interaction as fixed effects, using sum-to-zero contrasts. All models also included nested random effects for triads and participants. In the event of singularity warnings, we followed our preregistered pipeline and reduced the complexity of the random effects (Scandola & Tidoni, 2024). We report the results of the best-fit models, respectively. *P*-values for fixed effects were computed using Satterthwaite's approximation for degrees of freedom (Kuznetsova et al.,

2017). Post hoc pairwise comparisons of estimated marginal means were computed using emmeans (Lenth, 2024). In total, we ran models for (i) subjective AI literacy, (ii) objective AI literacy and its three subscales (AI literacy, technical literacy, programming literacy), and (iii) intrinsic motivation and its five subscales (interest/enjoyment, perceived competence, value/usefulness, pressure/tension, information sufficiency).

2.6.3 Students' and teachers' feedback (RQ4)

Quantitative feedback. Students' quantitative answers to the three post-test feedback items (enjoyment, difficulty, learning) were analyzed with three linear mixed models, each including the main effect of experimental group, and a random effect for triads. In the presence of singularity warnings, the random-effects structure was removed, and a linear model was fitted. These models were exploratory and not preregistered. Because our teachers' sample consisted of only two teachers, teachers' quantitative responses to the five post-test feedback items were used for descriptive purposes only. The ratings were averaged separately for Phase 1 and Phase 2 and visualized between the build and interaction conditions.

Qualitative feedback. Students' and teachers' qualitative feedback was analyzed with an inductive qualitative content analysis, a systematic approach for coding text data into meaningful categories (Krippendorff, 2018; Mayring, 2000), using MaxQDA (2024) and Python (Python Software Foundation, 2026). In total, 882 feedback segments were coded into six categories, derived from the data, which were deemed relevant to comparing participants' experiences in the building and interaction conditions. Specifically, these categories were: (i) robot building and assembly, (ii) programming and technical exploration, (iii) Marty interaction and activities, (iv) AI-related learning and communication, (v) learner/social engagement, and (vi) technical implementation and reliability. We used the same categories for participants' positive and negative feedback to summarize both what students and teachers mostly commented on and how positively or negatively these aspects were perceived. Responses not relevant to assessing different experiences between conditions, such as no-answer responses ($n = 113$), general positive feedback (e.g., liking everything; $n = 24$), general negative feedback (e.g., not finding anything difficult; $n = 145$), general dislike responses ($n = 3$), questionnaire-related comments ($n = 12$), and instructional-design comments ($n = 53$) were excluded from the analysis and are reported elsewhere (L. M. Leisten, Muss, et al., in prep.). The remaining data included 532 coded segments (students: $n = 456$; teachers: $n = 76$). Student feedback included 360 positive mentions (building: $n = 177$; interaction: $n = 183$) and 96 negative mentions (building: $n = 51$; interaction: $n = 45$). Teachers' feedback included 44 positive mentions (building: $n = 24$; interaction: $n = 20$) and 32 negative mentions (building: $n = 18$; interaction: $n = 14$).

We acknowledge that content analysis is a subjective process and, as such, is affected by the analysts' experiences and perspectives. To introduce multiple perspectives, we adopted a collaborative

approach in which the first author conducted the primary analysis in tandem with a student. Our interpretations were shaped by our team's diverse backgrounds in psychology, business engineering, neuroscience, cognitive sciences, and informatics, bringing together 3 to 20 years of research experience.

3 Results

3.1 Descriptive analyses

Mean values and standard deviations for each group at each time point are shown in **Table 2**. Correlations among the study measures are reported in the OSF project supplements (L. M. Leisten et al., 2025). Two participants failed both attention checks (T8), and nine participants failed one attention check (T1 = 2; T7 = 3; T8 = 4). As we implemented the attention checks only after T4 in the experimental group and did not preregister to exclude participants who failed them, these participants were still included in the analyses.

Table 2. Means and Standard deviations of study measures.

	T1 <i>M (SD)</i>	T4 <i>M (SD)</i>	T7 <i>M (SD)</i>	T8 <i>M (SD)</i>
Subjective AI literacy				
Interaction	5.7 (1.00)	6.17 (1.31)	6.66 (1.45)	
Building	5.86 (2.00)	5.58 (1.40)	7.01 (1.22)	
Objective digital literacy				
Interaction	5.79 (3.86)	6.82 (4.93)	8.65 (5.04)	8.86 (4.63)
Building	5.43 (3.17)	6.48 (3.61)	7.30 (3.37)	11.00 (3.52)
	Phase 1 M (SD)		Phase 2 M (SD)	
Intrinsic motivation				
Interaction		3.77 (0.65)	3.70 (0.57)	
Building		4.08 (0.68)	3.79 (0.81)	

3.2 Subjective AI Literacy (RQ1)

Likelihood ratio tests showed no evidence that condition ($\chi^2(1) = 0.85, p = .36$) or its interaction with time point ($\chi^2(2) = 0.49, p = .78$) predicted students' AI literacy (**Figure 4A**), indicating no evidence for different learning gains between conditions. However, including timepoint significantly improved model fit ($\chi^2(2) = 23.02, p < .001$). Compared to T1, subjective AI literacy scores significantly increased at T4 ($\beta = 0.56, SE = 0.22, p = .02$) and T7 ($\beta = 1.12, SE = 0.22, p < .001$). Post-hoc comparisons further showed a significant increase from T4 to T7 ($\beta = -0.56, SE = 0.23, p = .02$), indicating that children's subjective AI literacy increased from pre- to post-test.

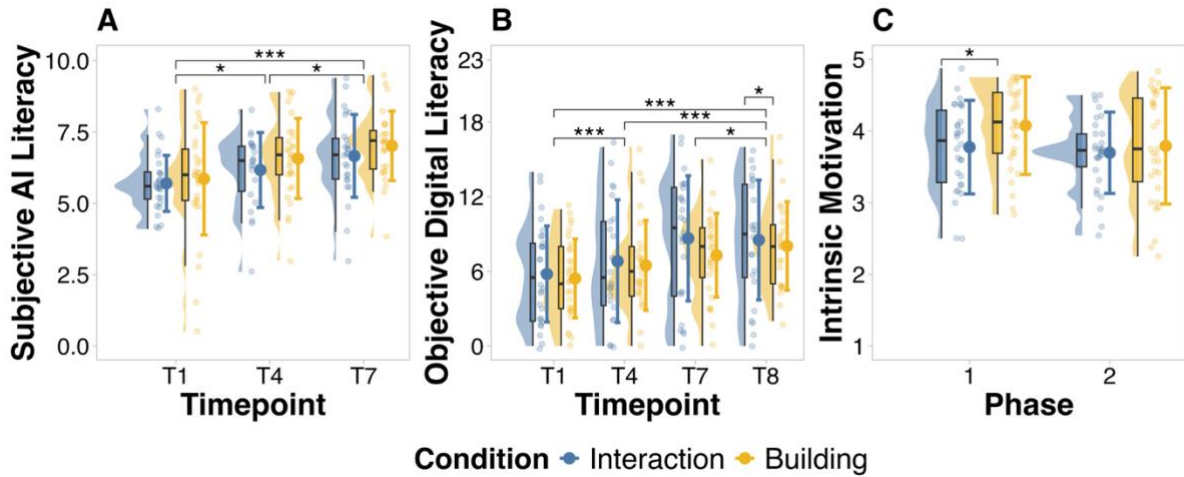


Figure 4. Changes of students' (A) subjective AI literacy, (B) objective digital literacy, and (C) intrinsic motivation, over time and by group. Each plot includes raw data points, means, and standard deviations.

3.3 Objective digital literacy (RQ2)

Similar to subjective AI literacy, likelihood ratio tests showed no evidence that condition ($\chi^2(1) = 0.14$, $p = .70$) or its interaction with time point ($\chi^2(3) = 1.07$, $p = .78$) predicted students' objective digital literacy (**Figure 4B**), providing no evidence for differences in learning between groups. Including timepoint significantly improved model fit ($\chi^2(3) = 28.105$, $p < .001$). Compared to T1, scores significantly increased to T7 ($\beta = 2.17$, $SE = 0.49$, $p < .001$). Children's objective literacy further increased in T8, as measured by the learning retention test conducted 7 weeks after the intervention ($\beta = 2.26$, $SE = 0.50$, $p < .001$). Children's objective literacy further increased in T8, as measured by the learning retention test conducted 7 weeks after the intervention. Post hoc comparisons indicated that this effect was mainly driven by the increase in the building group, which significantly differed from the interaction group ($\beta = 2.20$, $SE = 0.47$, $p < .001$). However, this effect may be due to an experimental error: children in the building group completed their retention-test survey without supervision, which may explain the unexpected increase relative to T7. We will thus not interpret the differences between groups but rather treat this effect as evidence of both groups' learning retention seven weeks after the intervention's completion. Pre-registered exploratory analyses of the individual subscales are available in the supplements (L. M. Leisten et al., 2025).

3.4 Intrinsic motivation (RQ3)

There was a significant interaction effect between condition and phase ($\beta = -0.05$, $SE = 0.02$, $p = .04$) on children's overall intrinsic motivation (**Figure 4C**). Post-hoc comparisons indicated that during phase 1, engagement was significantly higher in the building group than in the interaction group ($\beta = 0.35$, $SE = 0.16$, $p = .03$). This difference was no longer significant during phase 2 ($\beta = 0.16$, $SE = 0.17$,

$p = .33$), indicating that children in the building group enjoyed the activity more during phase 1 but not during later interactions.

The subscale analyses showed that the higher intrinsic motivation in the building group during phase 1 was primarily driven by differences on the *interest/enjoyment* and *value/usefulness* subscales. Specifically, for the *interest/enjoyment* subscale (**Figure 5A**), the results showed a significant main effect of condition ($\beta = 0.22$, $SE = 0.10$, $p = .01$), indicating higher interest and enjoyment in the building group than in the interaction group across both phases. Furthermore, the main effect of phase was significant ($\beta = -0.12$, $SE = 0.04$, $p = .002$), indicating greater enjoyment in the first phase than in the second for both groups. There was no significant interaction effect ($\beta = -0.05$, $SE = 0.04$, $p = .21$).

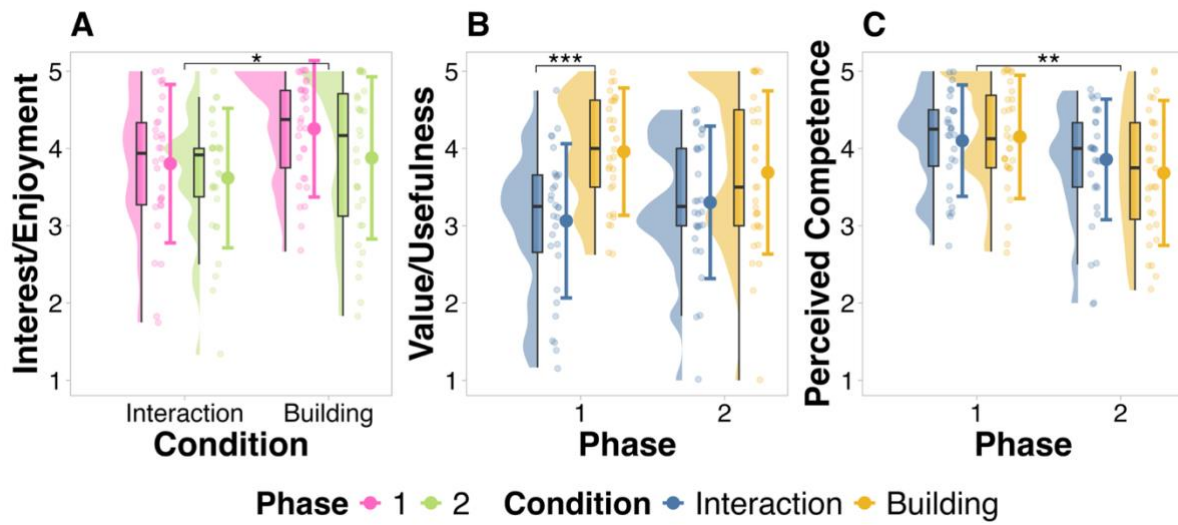


Figure 5. Changes of students' (A) interest/enjoyment, (B) perceived value/usefulness, (C) perceived competence, (D) information sufficiency, (E) and pressure, over time and by group. Each plot includes raw data points, means, and standard deviations.

For the *value/usefulness* subscale (**Figure 5B**), there was a significant interaction effect between phase and condition ($\beta = -0.13$, $SE = 0.03$, $p < .001$). Follow-up comparisons indicated a significant difference between the building and interaction groups in phase 1 ($\beta = 0.92$, $SE = 0.23$, $p < .001$), with the building group rating phase 1 significantly more useful than the interaction group. This difference disappeared in phase 2 ($\beta = 0.41$, $SE = 0.23$, $p = .09$).

We also found a main effect of phase ($\beta = -0.17$, $SE = 0.03$, $p < .001$) on students' *perceived competence* (**Figure 5C**), indicating a general decrease from phase 1 to phase 2 for both groups. No significant effects were found for condition ($\beta = -0.03$, $SE = 0.08$, $p = .74$) or their interaction ($\beta = -0.05$, $SE = 0.03$, $p = .09$). We did not find any significant effects regarding *information sufficiency* (p 's $> .53$) nor *pressure* (p 's $> .24$).

3.5 Students and teachers' feedback (RQ4)

3.5.1 Students' feedback

Quantitative. Children's ratings of the overall project regarding enjoyment, difficulty, and learning showed no significant differences between groups (p 's > .16; **Figure 6**). Relative to the 5-point scale, both groups rated the project as rather enjoyable ($M = 4.02$, $SD = 0.88$), moderately to rather difficult ($M = 3.48$, $SD = 0.97$), and reported learning neither a lot nor little ($M = 2.91$, $SD = 1.14$). These results are largely consistent with other measures in this study, which also showed no differences between groups regarding their learning or perceived pressure from the program. However, the non-significant enjoyment result does not align with the higher enjoyment/interest observed in the intrinsic motivation subscale. On that subscale, children in the building group reported higher enjoyment/interest on average throughout the project. We discuss possible explanations for this divergence in the discussion.

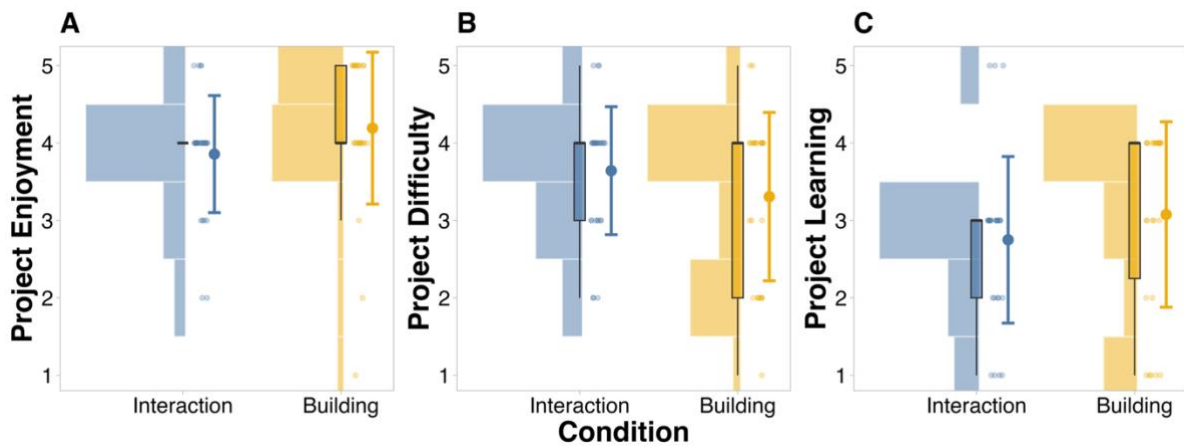


Figure 6. Students' ratings of the projects' A) enjoyment, B) difficulty, and C) perceived learning.

Each plot includes bar distributions, boxplots with median, raw data points, means, and standard deviations.

Qualitative Feedback. During Phase 1, the two conditions involved different activities: students in the build condition built Marty, whereas students in the interaction condition interacted with pre-built Martys through activities and games. Consequently, students' positive feedback reflected these different activity structures. Students in the build condition most often highlighted robot building and assembly ($n = 74$), including building Marty ($n = 49$), robot completion ($n = 18$), setup or unpacking ($n = 2$), and progress or working speed during the build ($n = 5$). This suggests that the build condition made construction, completion, and technical progress salient as part of students' experience. In contrast, positive feedback in the interaction condition during Phase 1 primarily focused on the immediate use of Marty for interaction and activities ($n = 94$). This included activities such as the penalty shootout ($n = 19$), the obstacle course ($n = 15$), Marty's dance ($n = 9$), and the escape game (n

= 10), as well as more general feedback around physical interaction ($n = 17$), Marty's control ($n = 8$), applying stickers or personalization ($n = 10$), and Marty's general appeal ($n = 6$).

A more informative pattern in Phase 1 was that children in the build condition mentioned programming and technical exploration ($n = 10$), whereas this category did not appear in the interaction condition. These comments mostly referred to testing or trying out Marty ($n = 7$) and programming Marty ($n = 3$), suggesting that some students may have started to experience Marty as a technical system rather than only as an interaction object. At the same time, AI-related learning and communication were mentioned more often in the interaction condition during Phase 1 (interaction: $n = 6$; build: $n = 1$), with students mainly commenting on learning or hearing about AI. This indicates that early positive feedback was not only shaped by the different activities but also by which content became salient in each condition.

As Phase 2 was designed with the same activities across conditions, differences in students' feedback during that phase may be especially informative. As expected, programming and technical exploration became equally prominent in both conditions (interaction: $n = 38$; build: $n = 38$). However, students in the build condition mentioned AI-related learning and communication more often than students in the interaction condition (build: $n = 28$; interaction: $n = 20$), whereas students in the interaction condition continued to refer more often to Marty interaction and activities (interaction: $n = 17$; build: $n = 1$). This pattern may suggest that Marty took on different roles across conditions. However, it should be interpreted cautiously, as the later scaffolded Marty/LLM activity worked as intended only in the build condition, whereas students in the interaction condition could not use the same Marty-based setup and instead engaged more generally with AI/LLM tools (see Limitations).

Learner and social engagement were mentioned slightly more often in the build condition across both phases (build: $n = 11$; interaction: $n = 8$). Such comments referred to the collaboration and working together in a group, as well as peer-related learning, such as receiving help from peers, helping others, or showing Marty's functions to peers. This suggests that the build condition may have supported not only technical engagement, but also moments of collaborative learning and possibly done so slightly more than the interaction condition.

Negative feedback was evenly distributed across categories and between conditions, suggesting that negative comments reflected specific implementation challenges rather than a general dislike of either condition. In the build condition, Phase 1 comments mainly reflected challenges with the build process, including slow progress and calibration difficulties ($n = 2$), unclear build guides ($n = 4$), and building difficulties ($n = 5$). However, the most frequent negative building-related category was students' wish for more time to build ($n = 6$). Thus, negative feedback about building was ambivalent: some children experienced difficulties, but others appeared frustrated because they wanted to continue building rather than because they disliked the activity. This suggests that the build condition was engaging, but required sufficient time, scaffolding, and technical support.

Learner and social engagement-related negative feedback appeared in both conditions, but differed across phases. In the interaction condition, most such negative comments appeared in Phase 1, while in the building group, more were made in Phase 2. The interaction commented mainly on peer behavior during the obstacle course activity ($n = 7$), including one student's complaint about perceived unfairness. In Phase 2, negative comments about group arrangements became more prominent ($n = 4$), and one student noted that not everybody was busy during the task. This may indicate group-level difficulties, but may also reflect that some later activities did not keep all students equally involved. In the build condition, fewer children reported negative comments about group-related difficulties ($n = 2$) during Phase 1, possibly indicating that the collaborative nature of the building activity helped mitigate conflict. During Phase 2, students in the building group similarly disliked group arrangements ($n = 5$). These comments, therefore, seem to reflect classroom and group-level implementation challenges that may be more pronounced in robot interaction than in robot building.

AI-related negative feedback differed between conditions and phases. In the interaction condition, Phase 1 comments mainly concerned difficulties with the introduction to AI and with understanding the concepts ($n = 4$), whereas in the building group, no negative comments were made on this topic. This may indicate that the robot-building activity better scaffolded the AI-related content. In Phase 2, negative AI-related feedback was mostly linked to the Marty/LLM interaction not working as intended ($n = 5$), while one child found writing to the LLM boring. This fits the later implementation issues in the interaction condition, where the Marty interface did not work reliably, and students partly used alternative LLM tools instead. In contrast, in the build condition, where the scaffolded Marty interaction worked, negative AI-related comments were less about access and more about the interaction design itself ($n = 3$).

Programming-related negative feedback was rare in Phase 1 but prevalent in Phase 2, with more mentions in the build condition. The comments reflected similar concerns across conditions: some students disliked programming, while others struggled to understand or implement the programming tasks. Lastly, technical implementation and reliability issues were present in both conditions across phases, although slightly fewer comments were made in the control group during Phase 1 (interaction: $n = 3$; building: $n = 6$). Overall, the feedback mainly included robot-related issues ($n = 14$) and other problems, such as programming or interface issues ($n = 7$). This illustrates how important reliable robots and tasks are for sustaining engagement, as issues may lead to frustration and risk the success of the deployment.

3.5.2 Teachers' evaluations

Quantitative Feedback. Teachers' ratings of students' engagement, learning, question-asking, and the difficulty of guiding them through the lessons are shown in **Figure 7**. In the following, we discuss visual patterns in the data that may indicate differences in experiences between groups from teachers'

perspectives. However, these data are descriptive and do not indicate formal differences. Means and standard deviations are available in the **Supplements** in our OSF project (L. M. Leisten et al., 2025).

Teachers rated students' engagement similarly high during Phase 1, but these ratings dropped in Phase 2 for the interaction condition (**Figure 7A**). A similar drop in Phase 2 was observed regarding relevant student questions (**Figure 7B**). The build condition was rated as more difficult to guide in Phase 1, both in terms of content (**Figure 7C**) and student behavior (**Figure 7D**). In Phase 2, both difficulty ratings were more similar across conditions. Students' learning was rated as moderately to highly supportive, and these ratings were relatively similar between conditions and Phases (**Figure 7E**).

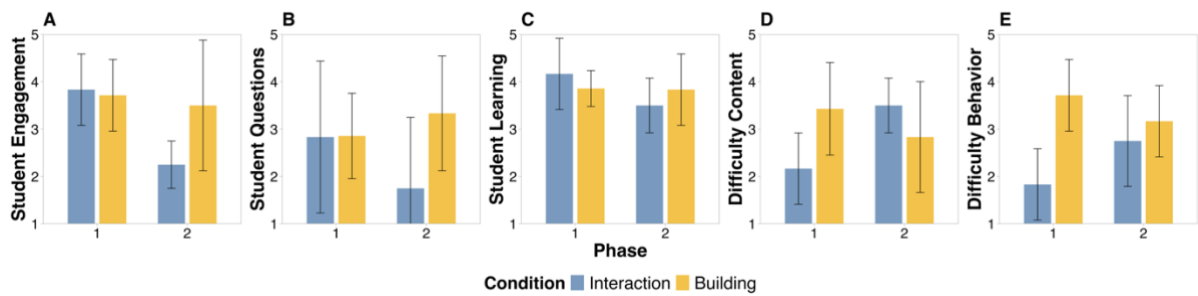


Figure 7. Descriptive teacher ratings of (A) student engagement, (B) relevant student questions, (C) perceived student learning, (D) difficulty guiding through content, (E) difficulty guiding student behavior, by phase and group. Each plot shows mean ratings averaged within phase and group, with standard deviations.

Qualitative Feedback. Heatmaps showing the occurrence of categories per condition and phase are available in the **Supplements** of our OSF project (L. M. Leisten et al., 2025). Positive teacher feedback in the build condition primarily referred to learner and social engagement ($n = 8$). This included group work ($n = 2$), with teachers noting that students worked well together and took turns, as well as students' motivation and interest ($n = 3$). Teachers also highlighted students' focused work ($n = 2$), both during the task itself and during the theoretical part, and one comment emphasized students' opportunities to explore Marty's features independently ($n = 1$). In contrast, robot building and assembly were mentioned less frequently ($n = 2$), suggesting that teachers perceived the value of building mainly through students' engagement rather than through the act of building itself.

In the interaction condition, positive teacher feedback during Phase 1 focused on Marty interaction and activities ($n = 5$), as expected, given that students interacted with pre-built Martys through activity formats. Teachers praised the activities, including the carefully prepared obstacle course, the Marty penalty shootout, which gave students more time to control Marty directly, and the escape game, which was particularly engaging. Across the project, positive feedback concerned programming and technical exploration ($n = 7$), with teachers highlighting opportunities for students to experiment ($n = 3$), program ($n = 3$), and work practically with Marty ($n = 1$).

Negative teacher feedback in the build condition primarily concerned robot building and assembly ($n = 8$). Teachers commented on students having difficulties building the robot ($n = 2$), uneven progress between groups ($n = 2$), and difficulties teaching or supporting the activity due to students' inexperience ($n = 4$). In two cases, this was explicitly connected to teachers feeling helpless or being overwhelmed. One teacher described being "*quite overwhelmed*" because there were "*so many new components to teach*" and they could not provide "*the usual level of support when difficulties arose*", illustrating the frustration that could arise when requiring teachers to support students without sufficient technical knowledge of the robot. The teacher further worried that accumulated difficulties for students could lead to "*too much frustration*" and even "*an aversion to the subject of programming*". This suggests that successful implementation requires not only student scaffolding but also sufficient teacher preparation to respond to arising difficulties. Additional negative comments referred to technical problems ($n = 2$), problems with the randomly assigned groups ($n = 1$), and distracted students ($n = 1$).

In the interaction condition, negative teacher feedback mostly concerned learner and social engagement ($n = 5$), especially difficulties with group work and student motivation. Teachers noted that collaboration was difficult in some groups because some students withdrew from the activity. Technical implementation and reliability were also mentioned ($n = 4$), mostly in relation to Martys not working as intended. These comments highlight the importance of teacher training, classroom dynamics, and reliable technical execution for long-term school deployments.

4 Discussion

In this classroom-based deployment study, we compared a robot-building and robot-interaction condition with a robot-interaction-only condition within a seven-week digital literacy curriculum to assess the extent to which adding robot building more effectively supports children's digital literacy and learning experiences. Our results indicate that both conditions led to improvements in students' subjective and objective knowledge, and that these learning gains persisted seven weeks after the intervention ended. Contrary to our pre-registered hypotheses, we found no evidence of group differences in objective or subjective learning outcomes (**RQ1 & RQ2**). However, children who built their own robots reported higher intrinsic motivation during the first phase, rated the curriculum as more useful, valuable, interesting, and enjoyable than students in the interaction group (**RQ3**). Students' and teachers' feedback indicated no formal evidence for differences in conditions, although qualitative feedback indicated nuanced trade-offs between engagement, learning benefits, and technical difficulties (**RQ4**). We discuss the implications of these findings below.

4.1 Subjective AI literacy and objective digital literacy (RQ1 & RQ2)

Contrary to our expectations (**H1 & H2**), we found no evidence that building robots alongside robot interaction is more effective than robot interaction alone in supporting children's subjective AI literacy

or objective digital literacy. Instead, students in both conditions improved their scores from pre- to mid- and post-test. That robot interaction can support children's learning across a number of different learning contexts is well established (Belpaeme et al., 2018), and our results contribute more data to this evidence base. What was less clear prior to this study was whether adding robot building would boost learning outcomes. Our findings suggest that neither approach has a demonstrable advantage over the other, but do support the viability of robot building as an alternative pedagogical approach within social robot-supported education.

While students' subjective AI literacy was already above the midpoint of the scale at pre-test ($M = 5.8/10$) and increased to a rather high ($M = 6.8/10$) self-rating over the course of the curriculum, objective digital literacy scores were below 25% at the start ($M = 5.6/23$) and reached only approximately 35% by the end of the intervention ($M = 8/23$). This gap could reflect that our objective literacy measure was too difficult to capture students' actual knowledge. However, since our digital literacy measure was mainly composed of AI literacy items, and children's objective AI literacy was as low as their overall digital literacy (see Supplements in L. M. Leisten et al., 2025), this could also indicate that subjective AI literacy is not a good proxy for objective knowledge. Possibly, children overestimated their (low) AI literacy skills, as is well established in a range of contexts by the Dunning-Kruger effect (Dunning, 2011; Jansen et al., 2021; Kruger & Dunning, 1999). As such, our study aligns with a growing body of evidence suggesting that subjective AI literacy is not a reliable proxy for objective knowledge (Chiu et al., 2024; Markus et al., 2025; Weber et al., 2023). Future work using age-appropriate objective measures is needed to determine how robustly this conclusion holds across different samples and instruments.

Beyond the comparison between conditions, our results contribute to the broader literature on HRI-supported learning in several ways. First, this study was among the first to examine the effects of a longer-term HRI intervention on adolescents' learning outcomes, as most prior work has focused on younger children or adults (Matheus et al., 2025). Studies with younger students have shown that sustained robot-supported learning can produce meaningful learning gains, for example, a scaffolding robot tutor supported 8–11-year-olds' math skills across three sessions (Ligthart et al., 2023), and an adaptive co-writing robot helped 5–8-year-olds overcome low handwriting confidence over repeated long sessions (Jacq et al., 2016). Our results extend this evidence to an older age group, showing that robots can meaningfully support adolescents' digital literacy education in long-term HRI and that the knowledge gained endures for at least seven weeks. Second, our work adds formal statistical evidence to a growing but methodologically limited literature on robot-supported AI literacy specifically. Williams and colleagues (2024) found pre-to-post increases in 9–11-year-olds' self-reported AI skills following interaction with the Doodlebot, and Wang and colleagues (2026) reported interview-based evidence of increased AI knowledge after children trained a social robot's algorithm, but neither study conducted formal significance testing due to small sample sizes. Third, our results show that learning gains from a seven-week HRI intervention remained stable at a follow-up assessment seven weeks post-

intervention. Evidence on the long-term retention of HRI-supported learning in adolescents is scarce (Matheus et al., 2025), and our findings suggest that robot-supported digital literacy gains may be durable. Future work should examine how retention varies across topics and age groups, when gains begin to fade, and whether longer interventions can yield even more enduring learning outcomes.

While the effectiveness of robot interaction for enhancing learning is increasingly documented across a range of learning contexts, evidence on robot building as a pedagogical approach in its own right remains scarce. Williams and colleagues (2019) showed that a combined robot-building and AI literacy activity improved theory-of-mind skills in 4–6-year-olds after one hour, suggesting that construction activities can carry pedagogical value beyond the technical domain. Kanda and colleagues (2012) investigated an eight-week deployment in which children built a LEGO Mindstorms robot while a Robovie robot served as a classroom teaching assistant. Although this design differs from ours (the robot children built was not the same social agent they interacted with), it offers a useful point of comparison by combining programmable robot construction with social robot interaction. Overall, children in that study improved their robotics knowledge, though it remains unclear whether those gains were attributable to the building activity, the social robot, or their combination. Our results add to this evidence base by demonstrating that robot building can effectively support both objective knowledge and subjective learning experience. Additionally, the observation of stable gains seven weeks after lesson delivery demonstrates that these gains are not quickly lost. However, future work is needed to further investigate differences between robot building and robot interaction on other factors of children’s classroom interactions and the learning process. For example, robot-building programs may also serve as tools to support teachers’ digital literacy and confidence with technology, and may increase students’ perceptions of their teachers as digitally competent, factors that are important for successful learning support (Ali et al., 2020; San-Martín et al., 2021; Turulja et al., 2020).

4.2 Intrinsic motivation (RQ3)

In line with **H3**, our results indicate that building robots was perceived as more enjoyable and useful than interacting with robots. This finding is particularly noteworthy because both conditions were designed to be engaging, and the interaction-only condition included several game-like activities, such as soccer tournaments, obstacle-course design and racing, and an escape game. By comparison, students in the building condition sometimes encountered practical difficulties, including assembly problems, limited materials, and waiting periods after finishing earlier than their peers. Despite these challenges, students still perceived robot-building as more useful and, overall, reported higher intrinsic motivation than students in the interaction-only condition.

An implication of this finding is that robot-building may help cultivate forms of learning-related resilience. Here, we draw on productive struggle theory, which posits that struggle is a necessary component of learning and understanding (Warshauer, 2015; Young et al., 2023). Unlike interaction

activities that primarily emphasize use, construction tasks require students to cope with setbacks, delays, and imperfect outcomes while continuing to work toward a functioning artifact. In this sense, robot-building may provide structured opportunities for productive struggle and for practicing perseverance and frustration tolerance (Bolyard et al., 2023), although these were not directly measured in the present study.

According to the self-determination theory (Ryan & Deci, 2000), intrinsic motivation increases when people perceive an activity as meaningful and aligned with their own goals. When learners internalize the value of an activity, they engage more deeply and sustain effort not because they are told to, but because the activity feels important. Building a robot may have strengthened this sense of personal meaning in ways that interacting with a pre-assembled robot did not: students could see, hold, and demonstrate a tangible product of their effort, making the connection between their learning and its outcome concrete and direct. This explanation is in line with prior work on the IKEA effect, in which building artifacts is linked to feelings of ownership, attachment, and intention to use such products (Norton et al., 2012; Pelled et al., 2026; Sun & Sundar, 2016; Tsumura & Yamada, 2025; Williams et al., 2019). This explanation may explain why the building group rated the curriculum as more useful and important, and why the perceived value of the building activity was related to their higher overall intrinsic motivation.

The building group also rated the entire curriculum as more enjoyable regardless of the phase they were in, indicating that robot building affects motivation not only during the build itself but also during later interactions with the finished robot. One possible explanation, which was already discussed above, is that building fostered a stronger sense of ownership and personal investment (Groom et al., 2009; Sun & Sundar, 2016; Tsumura & Yamada, 2025; Williams et al., 2019), such that later interactions were experienced not simply as using a robot, but as engaging with something the students had helped create. This is supported by Tsumura and Yamada's (2025) work, which found that assembling a robot can increase adults' empathy towards it. In our sample, construction may have created a more continuous relationship between students and the robot, making subsequent activities feel more meaningful and personally relevant. This finding is important because it suggests that robot-building may support more enjoyable interactions and possibly sustained engagement with social robots over time, which remains a persistent challenge in educational HRI (Barakova et al., 2023; Leite et al., 2013; Tisza et al., 2025).

However, this motivational advantage did not translate into greater cognitive learning gains. Intrinsic motivation is generally considered beneficial for learning (Froiland & Worrell, 2016; Wulf & Lewthwaite, 2016), and this relationship has also been observed in children's learning interactions with social robots (Ramachandran et al., 2019). At the same time, affective gains in HRI do not consistently translate into cognitive gains, as noted in prior work (Belpaeme et al., 2018), possibly because learning outcomes depend not only on motivation but also on the structure of the learning activity itself. Our results add to this literature by suggesting that higher intrinsic motivation during a digital literacy

curriculum does not necessarily lead to greater gains in objective digital literacy. One lens for interpreting this pattern is Chi's (2009) taxonomy of active, constructive, and interactive learning. Within this framework, interactive learning is expected to yield stronger learning outcomes than constructive or active engagement alone. In our study, students in both conditions worked collaboratively and therefore had opportunities for interactive learning, while students in the building condition additionally engaged in constructive activities through assembling the robot. The absence of group differences in cognitive learning may therefore suggest that the curriculum's shared interactive features were already sufficiently powerful to support learning in both conditions, reducing the likelihood that the additional motivational benefits of building would yield measurable cognitive advantages. That said, the motivational benefits of robot-building need not be considered separately from cognitive goals: educators could use the increased sense of value and engagement associated with building activities to support particularly challenging or abstract content, where additional motivational support may be especially beneficial.

4.3 Students' and teachers' project feedback (RQ4)

Children's post-test feedback regarding overall project enjoyment did not differ significantly between groups, standing in contrast to the higher enjoyment and interest scores reported on the intrinsic motivation subscale. There could be multiple explanations for this. First, it could be due to the subscale's two-item structure, which included both interest and general enjoyment. Second, it may be due to temporal dynamics. Due to the recency effect (Atkinson & Shiffrin, 1968; Luebbbers et al., 2024), individuals tend to recall more recent events more vividly than earlier ones. Because children's activities in week 7 were (almost) identical across groups, and because the differences in enjoyment between groups were less pronounced during Phase 2, these final sessions may have had more influence on their retrospective evaluation of the entire project. Ultimately, this finding may suggest that while the positive impact of robot assembly on children's engagement is particularly pronounced during the active building phase, it may not fundamentally alter their long-term perception of the project as a whole, at least regarding their enjoyment.

Students' qualitative feedback indicated that collaborative robot building may be protective against group conflicts, helps scaffold AI-related learning content, and shifts students' focus towards Marty's role in supporting AI-related learning rather than on its role as an interaction partner. These findings, while tentative, may suggest that robot-building might change the role a robot assumes in a classroom, and whether that role is focused on the robot interaction itself or instead promotes learning activities and collaboration amongst students. However, these differences did not seem to transfer to formal learning differences between conditions, and more work is needed to further assess these effects.

Quantitative and qualitative feedback of teachers indicated practical difficulties in the implementation of the robot-building, mainly due to technical difficulties and content overload.

However, teachers evaluated the project as quite positive for children's learning and engagement, and while those ratings dropped in the interaction groups, they remained stable in the building group. This may indicate a slight advantage of robot building over robot interaction if technical difficulties could be solved. However, due to the small number of teachers, we cannot derive any formal differences from this data and further work is needed to validate these results.

4.4 Broader implications for educational social robotics

Our findings highlight the feasibility of deploying low-cost social robots in classrooms and integrating them into an existing curriculum over an extended period. Our results align with prior work demonstrating the value of long-term, in-the-wild robot deployments for children's learning, and extend most such deployments by additionally providing evidence for objective and subjective learning gains (Davison et al., 2020; Kanda et al., 2007; Kupaei et al., 2026; Matheus et al., 2025). Furthermore, our results suggest that having students build the robot themselves is at least as effective for learning as having them interact with a pre-assembled version, meaning the additional effort and complexity of a building-based approach does not come at an educational cost, and can bring about demonstrable gains in motivation and investment in learning, even if cognitive/performance gains are not recorded. Given that DIY robotic toolkits are significantly less costly and less technically complex than commonly used off-the-shelf systems such as the NAO or Pepper robots (*Aldebaran Robotics*, 2023; Alves-Oliveira et al., 2019; Papakostas et al., 2018; Tzampazaki et al., 2024; Woo et al., 2021), building-based approaches may offer a more accessible entry point to robot-supported education for all schools regardless of financial resources.

Beyond their practical advantages, building-based approaches may also require us to reconsider how social robots are conceptually positioned within educational settings. Our work provides an example in which a social robot neither adopts the role of a social agent nor that of a programmable tool, but rather combines both, along with a new role as a buildable agent. This exploration of new roles aligns with a recent call to broaden the functions of social robots in educational settings, moving away from typical roles such as tutor, peer, or teacher towards more diverse and participatory ones (Rohlfing et al., 2022). Repositioning the robot as something students actively construct, rather than as a system imposed upon them, may also help address the persistent challenge of stakeholder acceptance in educational HRI (Barakova et al., 2023; L. M. Leisten, Graw, et al., 2026; Smakman, Konijn, et al., 2021; Smakman, Vogt, et al., 2021). Research supports this: higher levels of user involvement in designing educational robots have been associated with greater intention to use them (Weidmann et al., 2024), and participatory design approaches have been shown to increase system transparency and trust (Gebelli et al., 2024). Involving students and teachers in the building process could thus give the school community genuine agency over the technology, engaging them in decisions about its features and design, an aspect identified as critical for the sustainable implementation of robots in schools (L. M.

Leisten, Graw, et al., 2026; Tisza et al., 2025). Platforms that support this kind of user involvement already exist: the DIY robot Blossom (Suguitan & Hoffman, 2019) allows users to customize its physical shell, YOLO (Alves-Oliveira et al., 2017) allows users to program its movements through puppeteering, and Reachy Mini (Pollen Robotics & Hugging Face, 2026) allows the community to program and share applications for it. The existence of such platforms suggests that the infrastructure for participatory, building-based approaches is already in place; what is needed now is systematic research into how these designs can be implemented and scaled across diverse educational contexts.

4.5 Limitations

As with any empirical HRI work, and especially with an in-the-wild deployment, it is vital to consider the limitations of our present approach, as well as constraints on generality more broadly. Although students showed significant gains in objective digital literacy, absolute scores remained low throughout the intervention, with students achieving only approximately 35% correct on the post-test. This reflects, at least in part, a measurement challenge: no age-appropriate, objective measures of AI literacy currently exist that are independent of any specific pre-existing curriculum or that assume students' prior AI literacy. Consequently, the instrument used in this study was adapted, in part, from a measure originally developed for adults. Although we selected mainly easy items and adapted the phrasing, feedback from teachers indicated that the content of several items remained too difficult for the age group. Scores may thus underestimate students' actual knowledge, and the practical significance of our learning effects should be interpreted cautiously. Future work should develop validated, age-appropriate objective measures of AI and digital literacy, adapted to different stages of students' digital education, which varies across countries (Sajadieh et al., 2026). Data quality in school-based research posed an additional challenge. Although we included attention checks and emphasized careful completion, survey completion times varied across measurement points, suggesting that response effort was not consistent throughout the study. Future studies should explore alternative strategies to protect against careless responding, such as linking assessments to in-class grades.

Second, several challenges arose in deployment and experimental design, as is often the case in long-term, in-the-wild studies (Woo et al., 2021). Technical issues with the robot's language model during the interaction-condition sessions meant that the originally planned verbal interactions with Marty in week 7 could not take place and were instead replaced with additional programming instructions. This introduced unintended experiential variance between conditions, which is also reflected visually (though not statistically significant) in higher programming knowledge in the interaction condition (see Supplements in L. M. Leisten et al., 2025). Additionally, curriculum pacing varied across classrooms, with some sessions not covering all intended content, resulting in adjustments between sessions and conditions. These deviations, while typical of real-world classroom settings, possibly introduced noise into the data. The physical layout of the deployment, with two adjacent classrooms between which

students could potentially move or communicate, required that both classes receive the same conditions at the same time. Because teachers taught the building condition first, they were more familiar with the curriculum by the time they delivered the interaction condition, which may have improved instructional quality for the interaction group. Teachers were also not fully blinded to the research questions. As active participants in the curriculum design, they were necessarily informed about both conditions and may have held expectations that influenced their teaching. Teacher affect presents an additional concern. In a within-teacher design, practical lessons learned during the first round of instruction, such as timing issues or content difficulty, could not be incorporated into the second round without compromising consistency between conditions. This meant that teachers had to repeat the same difficulties in the interaction group phase, leading to frustration and reduced enthusiasm among at least one teacher. Future studies should consider counterbalancing the order of conditions across teachers and using fully standardized delivery protocols. However, these are difficult to implement when teachers, rather than researchers, deliver the content. Additionally, while Marty served as an accessible and affordable DIY platform for this study, it is not fully open-source, which limits replicability and community-driven adaptation. Future research should examine the potential of fully open-source platforms, such as Blossom (Suguitan & Hoffman, 2019), to support children's digital literacy. Finally, at the last measurement point (T8), an administrative oversight led the building group to complete their survey without teacher supervision. Consequently, we cannot rule out that students consulted each other or external resources. This is reflected in an increase in objective digital literacy scores between T7 and T8 in the building group only, which is why we did not interpret the between-group difference at that time point.

4.5.1 Constraints on generality

In line with Simons and colleagues (2017), we discuss the constraints of generality in this section by identifying our target population and highlighting its limits of generalizability. We targeted secondary students in a German educational setting with our curriculum, and the characteristics of our sample restrict the extent to which findings can be generalized beyond this specific context. Our sample was drawn from a single school in Germany with relatively high socioeconomic status and showed a significant gender imbalance, with more boys than girls participating. This is particularly relevant given that stereotypical gender norms continue to shape unequal access to STEM and robotics activities and careers, with girls remaining systematically underrepresented in these domains (Hernández-Pérez et al., 2024; Master, 2021). Whether the effects observed here hold for girls, for students from lower socioeconomic backgrounds, or for students in different national or cultural contexts remains an open question. More broadly, our findings are specific to this particular robot, curriculum, and deployment context, and caution is warranted in generalizing to other platforms, age groups, or educational settings (Belpaeme et al., 2018; Cross & Ramsey, 2021). We are currently extending this line of work with the

Blossom robot in a remote Australian context with younger children, providing additional context to these findings across varying conditions.

4.6 Conclusion

This study compared robot building and robot interaction as pedagogical approaches within a seven-week digital literacy curriculum, examining the extent to which building would more effectively support children's AI and digital literacy, as well as their intrinsic motivation. Our results show that both approaches led to subjective and objective knowledge gains, and that these gains remained stable seven weeks after the intervention ended, providing some of the first longitudinal evidence that HRI-based learning interventions can produce durable effects in adolescents. While no differences in learning outcomes were found between conditions, children in the robot-building condition reported higher intrinsic motivation, rating the curriculum as more useful, valuable, and enjoyable. Together, these findings suggest that robot building can be useful for supporting digital literacy to a comparable extent to interacting only with a pre-built robot, and also carries motivational advantages that may be particularly valuable for sustaining long-term engagement with educational technology.

These results have several implications. First, DIY robotic toolkits represent an accessible and cost-effective entry point into robot-supported education that need not come at an educational cost. Second, combining construction activities with more complex or abstract content could help educators use motivation to support more challenging learning experiences. More broadly, our findings point to the value of repositioning social robots in educational settings, not merely as tools to interact with, but as social agents to construct.

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